PROGRAMMING WITH PYTHON

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Computer Applications

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Python Basics and Standard Library
Introduction to Python Programming

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Introduction

- Python is a programming language created by Guido van Rossum.
- It is a powerful high-level, object-oriented programming language.
- Python is a general-purpose language.

Advantages of Python

- **A simple language which is easier to learn.**
  - Python has a very simple and elegant syntax.
- **Free and open-source**
  - Freely use and distribute Python, even for commercial use.
- **Portability**
  - Python programs can be moved from one platform to another, and run without any changes

Advantages of Python

- **Extensible and Embeddable**
  - Pieces of C/C++ or other languages can be combined with Python code.
- **It is a high-level, interpreted language**
  - When Python code is run, it automatically converts the code to the language the computer understands.

Advantages of Python

- Large standard libraries to solve common tasks
  - Python has a number of standard libraries which makes life of a programmer much easier since you don't have to write all the code yourself.
- Object-oriented
  - Everything in Python is an object.
  - Object oriented programming (OOP) helps to solve a complex problem intuitively.

Program Translation

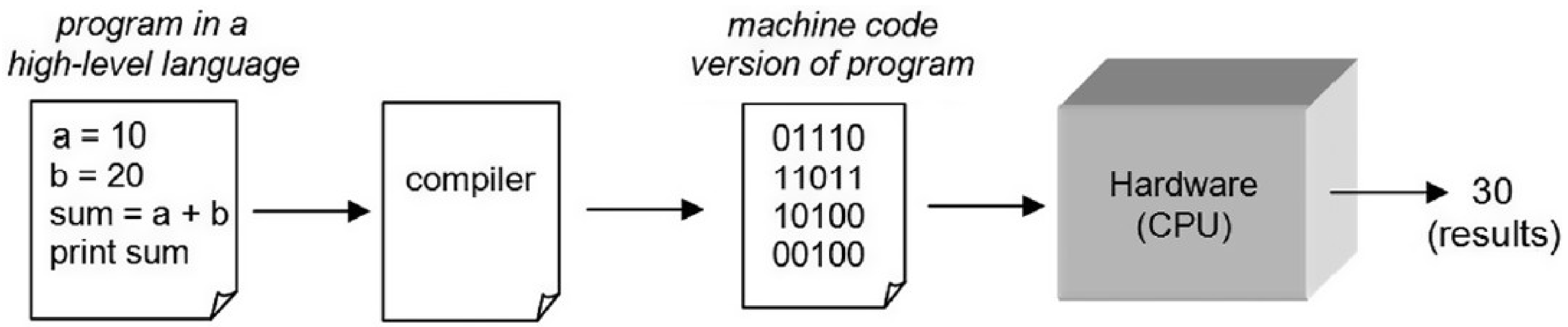
- Writing programs at “low level” of 0's and 1's is tedious and error-prone.
- Therefore, most programs are written in a “high-level” programming language such as Python.
- Since the instructions of such programs are not in machine code that a CPU can execute, a translator program must be used.

Program Translation

- There are two fundamental types of translators:
  - Compiler
    - translates programs into machine code to be executed by the CPU
  - Interpreter
    - executes instructions in place of the CPU

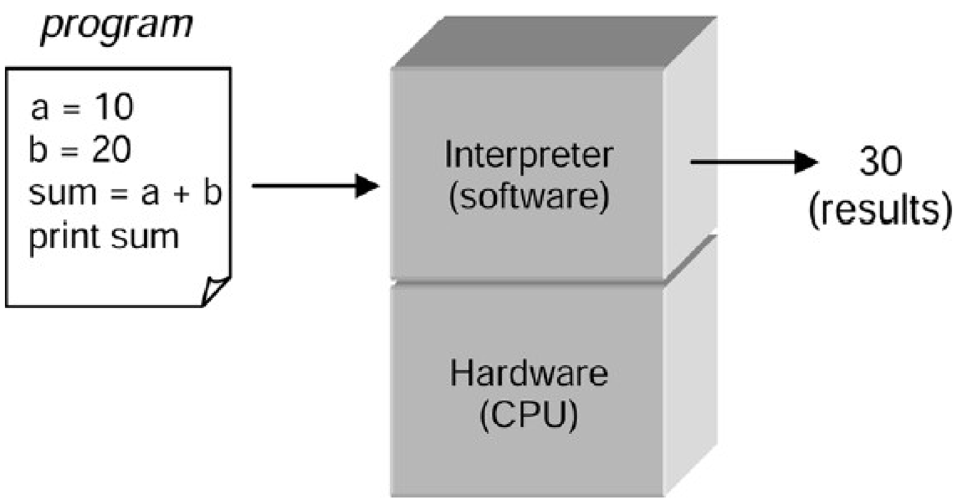
Compiler

- Compiled programs generally execute faster than interpreted programs



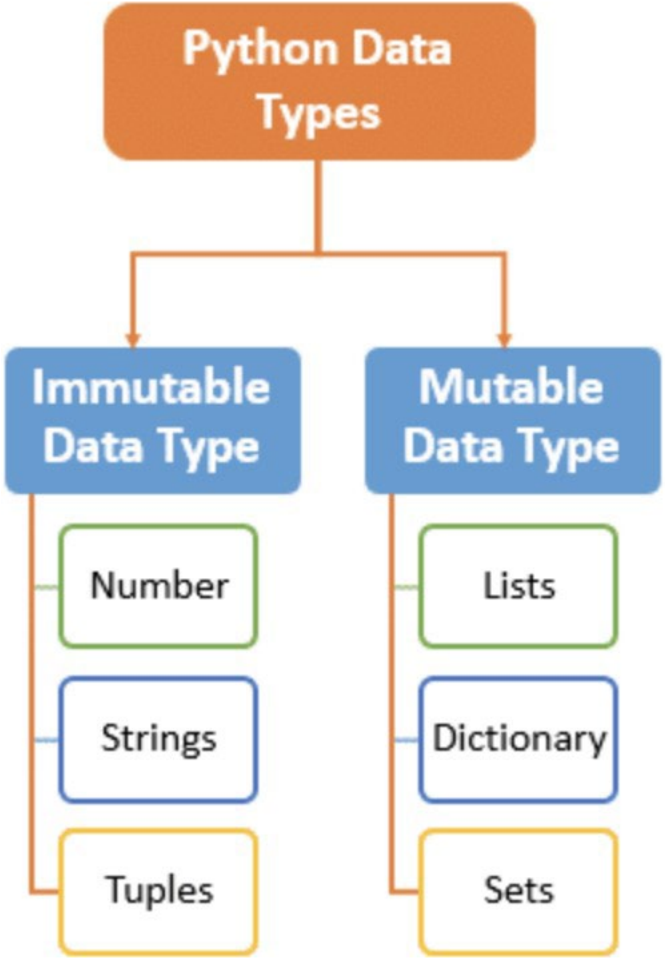
Interpreter

- An interpreter can immediately execute instructions as they are entered.
- This is referred to as ***interactive mode***.
- Python is executed by an interpreter.



Data Types

- Numbers
- List
- Tuple
- Strings
- Set
- Dictionary



- Numbers
  - Integers – int
    - Integers can be of any length, it is only limited by the memory available.
  - Floating point numbers – float
    - A floating point number is accurate up to 15 decimal places.
    - Integer and floating points are separated by decimal points. 1 is integer, 1.0 is floating point number.

- Numbers
  - Complex numbers - complex
    - Complex numbers are written in the form, x + yj, where x is the real part and y is the imaginary part.

```
x = 11     #int
y = 2.8    #float
z = 1+3j   #complex

#convert from int to float:
a = float(x)

#convert from float to int:
b = int(y)

#convert from int to complex:
c = complex(x)
```

- Numbers of higher type cannot be converted to lower types

```
#convert from complex to int:
d = int(z)

#convert from complex to float:
d = float(z)
```

- Strings
  - String is sequence of Unicode characters.
  - Single quotes or double quotes can be used to represent strings.
  - Multi-line strings can be denoted using triple quotes, ''' or """.
  - Strings can be indexed - often synonymously called subscripted as well.
  - Similar to C, the first character of a string has the index 0.

- Some methods on string
  - Count()
    - Returns the number of times a specified value occurs in a string
  - Find()
    - Searches the string for a specified value and returns the position of where it was found.

- Some methods on string
  - Partition()
    - Returns a tuple where the string is parted into three parts – before match, match and after match
  - Index()
    - Searches the string for a specified value and returns the position of where it was found

- Some methods on string
  - Split()
    - Splits the string at the specified separator, and returns a list

- Lists
  - List is an ordered sequence of items.
  - It is one of the most used datatype in Python and is very flexible.
  - All the items in a list do not need to be of the same type.
  - Declaring a list is pretty straight forward.
  - Items separated by commas are enclosed within brackets [ ].

- List can be created using [] or list() constructor method.

```
#List creation
l=[1,'two', 3.5,[2,3,4],{2,3,4}]
print(l)
print(type(l))
#print using index
for i in l:
    print(i)
#List creation
my_string = "Python"
my_list = list(my_string)
print(my_list)
```

- Methods on list
  - Append()
    - Adds an element at the end of the list
  - Insert()
    - Adds an element at the specified position
  - Pop()
    - Removes the element at the specified position

- Remove()
  - Removes the first item with the specified value
- Reverse()
  - Reverses the order of the list
- Sort()
  - Sorts the list

- Tuples
  - Tuple is an ordered sequence of items.
  - Tuples are immutable.
  - Tuples once created cannot be modified.
  - Tuples are used to write-protect data and are usually faster than list as it cannot change dynamically.
  - It is defined within parentheses () where items are separated by commas.

- Tuples can be created using () and tuple constructor.
- Methods on tuple
    - count()
        - Returns the number of times a specified value occurs in a tuple
    - index()
        - Searches the tuple for a specified value and returns the position of where it was found

- Sets

  - Set is an unordered collection of unique items.

  - A set is a collection which is unordered, unchangeable, and unindexed.

    - Unordered means that the items in a set do not have a defined order.

    - Set items can appear in a different order every time they are used and cannot be referred to by index or key.

  - Set is defined by values separated by comma inside braces { }.

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- Sets can be created using {} or the set constructor.
- Even if duplicate values are written in a set, they don't get stored in a set.

- Methods
  - add()
    - Adds an element to the set
  - intersection()
    - Returns a set, that is the intersection of two or more sets

- pop()
  - Removes an element from the set
- remove()
  - Removes the specified element
- union()
  - Return a set containing the union of sets

- Dictionary

  - A dictionary is a collection which is unordered, changeable and indexed.

  - Dictionaries are written with curly brackets.

  - They have keys and values.

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  - A dictionary is a collection which is unordered, changeable and indexed.
  - Dictionaries are written with curly brackets.
  - They have keys and values.

- Methods
  - get()
    - Returns the value of the specified key
  - items()
    - Returns a list containing a tuple for each key value pair
  - keys()
    - Returns a list containing the dictionary's keys

- Methods
  - pop()
    - Removes the element with the specified key
  - popitem()
    - Removes the last inserted key-value pair
  - values()
    - Returns a list of all the values in the dictionary

Recap

- Introduction
- Advantages of Python
- Data Types

THANK YOU

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